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141. Linked List Cycle

Solved

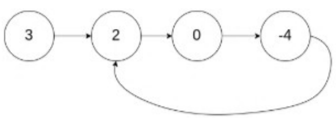
Easy Topics Companies

Given `head`, the head of a linked list, determine if the linked list has a cycle in it.

There is a cycle in a linked list if there is some node in the list that can be reached again by continuously following the `next` pointer. Internally, `pos` is used to denote the index of the node that tail's `next` pointer is connected to. **Note that `pos` is not passed as a parameter.**

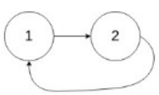
Return `true` *if there is a cycle in the linked list*. Otherwise, return `false`.

Example 1:




```
Input: head = [3,2,0,-4], pos = 1
Output: true
Explanation: There is a cycle in the linked list, where the tail connects to the 1st node (0-indexed).
```

Example 2:




```
Input: head = [1,2], pos = 0
Output: true
Explanation: There is a cycle in the linked list, where the tail connects to the 0th node.
```

Example 3:


```
Input: head = [1], pos = -1
```

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Code

C Auto

```c
/**
 * Definition for singly-linked list.
 * struct ListNode {
 *     int val;
 *     struct ListNode *next;
 * };
 */
bool hasCycle(struct ListNode *head) {
    if (head == NULL || head->next == NULL) {
        return false;
    }

    struct ListNode *C1 = head;
    struct ListNode *C2= head;

    while (C2 != NULL && C2->next != NULL) {
        C1 = C1->next;
        C2 = C2->next->next;

        if (C1 == C2) {
            return true;
        }
    }

    return false;
}
```

Saved Ln 22, Col 10

Testcase | Test Result

Accepted Runtime: 3 ms

Case 1 Case 2 Case 3

Input

head =

[1]

pos =

-1

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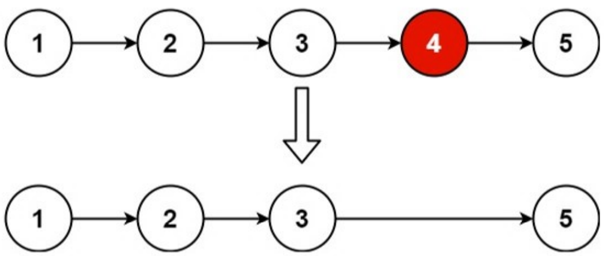
19. Remove Nth Node From End of List

Solved

Medium | Topics | Companies | Hint

Given the `head` of a linked list, remove the n^{th} node from the end of the list and return its head.

Example 1:



Input: head = [1,2,3,4,5], n = 2
Output: [1,2,3,5]

Example 2:

Input: head = [1], n = 1
Output: []

Example 3:

Input: head = [1,2], n = 1
Output: [1]

Constraints:

- The number of nodes in the list is `sz`.
- `1 <= sz <= 30`

20.8K | 319 | 0 Online

Code

C | Auto

```c
 * Definition for singly-linked list.
 * struct ListNode {
 *     int val;
 *     struct ListNode *next;
 * };
 */
struct ListNode* removeNthFromEnd(struct ListNode* head, int n) {
    struct ListNode list;
    list.next = head;

    struct ListNode* fast = &list;
    struct ListNode* slow = &list;

    for (int i = 0; i <= n; i++) {
        fast = fast->next;
    }

    while (fast != NULL) {
        fast = fast->next;
        slow = slow->next;
    }

    struct ListNode* nodeToDelete = slow->next;
    slow->next = slow->next->next;

    free(nodeToDelete);

    return list.next;
}
```

Saved | Ln 23, Col 1

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2 | Case 3

Input

head =
[1,2,3,4,5]

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